Splitting a middle generator in the symmetric group

Human Compression project

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Abstract

We apply the common supported-transposition theorem to the complete principal rank-augmentation statement of De Saedeleer–Leemans–Mulpas, in its specified author edition. The original four- and five-point cases are proved with their original hypotheses. The six-point assertion is false as printed; we give an exact replacement and bind a complete counterexample. We also verify all three finite examples used by the source to motivate its additional assumptions. The introductory census is retained as background data rather than asserted as a new enumeration.

1 The original statement and the corrected conclusion

The source is the nineteen-page author edition arXiv:2212.12563v1, 23 December 2022 [DLM]. Its Theorem 4.1 starts on physical page 5 and continues on page 6. We retain its distinction between the natural permutation group S_n and the particular chosen marking. Write its original rank-three C-marking as (a, q, c). Choose a transposition t from the disjoint cycle expression of q, with neither endpoint moved by c, and write $q = tb$. The proposed new marking is (t, a, b, c). Put

$$B = \langle a, b\rangle, \quad H = \langle t, a, b\rangle, \quad K = \langle a, b, c\rangle,$$

and let O be the H-orbit containing the endpoints of t.

Theorem 1. *Assume the complete hypotheses of [DLM, Theorem 4.1]: $n \geq 5$; (a, q, c) is a rank-three C-marking of the natural S_n; q is not a transposition; the chosen q-edge is not adjacent to a c-edge; and every B-orbit has size at most three. Then the following hold.*

1. *If $|O| = 4$ and K acts imprimitively on each of its orbits containing more than one B-triple, the new marking is C.*
2. *If $|O| = 5$, a, b, c are even, and the same imprimitivity condition holds, the new marking is C.*
3. *If $|O| = 6$, let N be the group generated by the three matching transpositions between its two marked B-triples. The new marking is C if and only if $N \cap K = 1$. It is C, in particular, when those two triples belong to distinct K-orbits. Without an additional condition, it need not be C.*

Proof. The new marking generates the natural S_n, since it generates the original marks $a, q = tb, c$. The chosen transposition commutes with both b and c, and the old string condition gives $ac = ca$. The local component argument in the common paper gives exactly its four-, five- and six-point paths; in particular the four marks are distinct nonidentity involutions. Its two-end lemma therefore applies. The common block-and-parity theorem proves (1) and (2). Its theorem on the six-point action proves the exact criterion and sufficient condition in (3). The frozen eighteen-point counterexample, recalled below, disproves the unrestricted assertion. Thus the result preserves both valid source cases and replaces its false third case explicitly. □

The common proof also establishes the elementary dihedral description in source Lemma 5.2, both end C-markings in Sections 5.1–5.2, and both conclusions of its block Lemma 5.3. In particular, no failure of the last intersection is being inferred merely from an error in the source's proof: the next section gives a group element in the wrong intersection.

2 The actual six-point obstruction

On eighteen points use

$$\begin{aligned} a &= (1\,2)(5\,6)(7\,8)(11\,12)(13\,14)(17\,18),\\ b &= (2\,3)(4\,5)(7\,9)(10\,11)(14\,15)(16\,17),\\ c &= (3\,4)(5\,7)(6\,8)(9\,10)(11\,13)(12\,14)(15\,16),\\ t &= (1\,18), \qquad q = tb. \end{aligned}$$

The complete written proof in the separately frozen correction establishes that (a, q, c) is an irreducible C-marking of the natural S_{18}, of type $[6, 90]$. It supplies the whole permutations, a seventeen-edge conjugate-transposition tree proving generation, and the exact dihedral intersection proving all original parabolic identities. Its input hypotheses are part of this application's full dependency boundary, not an unpriced citation to a short diagnostic.

The six B-orbits are consecutive triples. The H-orbit containing t is $\{1, 2, 3, 16, 17, 18\}$, of size six. With products acting on the right, put $w = a(bc)^2ba(cb)^8a$. Direct multiplication gives

$$(bc)^{45} = (6\,8), \qquad 6^w = 1, \quad 8^w = 18, \qquad ((bc)^{45})^w = t.$$

Thus $t \in K$ and $N \cap K \neq 1$. The label sets $\{t\}$ and $\{a, b, c\}$ are disjoint, so this is an actual failure of the intersection property. All thirteen statements in the separate Lean supplement concern these literal output permutations and generated subgroups. They do not formalize the old C-marking, natural symmetric group identification, or source theorem hypotheses, which retain their complete written proofs.

3 The three motivating examples

Section 6 of [DLM], physical pages 17–18, supplies examples showing why the additional assumptions in the first two cases matter. We keep its exact point labels. The original triples (a, q, c) and the split transpositions are as follows:

$$\begin{aligned} n = 9:\quad & a = (2\,3)(6\,9)(7\,8), && q = (1\,2)(3\,4)(5\,6),\\ & c = (4\,5)(6\,7)(8\,9), && t = (1\,2);\\ n = 10:\quad & a = (1\,2)(3\,4)(9\,10), && q = (2\,3)(4\,5)(6\,7)(8\,9),\\ & c = (5\,6)(7\,8), && t = (2\,3);\\ n = 13:\quad & a = (1\,2)(3\,4)(7\,9)(8\,10), && q = (2\,3)(4\,5)(6\,7)(10\,11)(12\,13),\\ & c = (5\,6)(7\,8)(9\,10)(11\,12), && t = (2\,3). \end{aligned}$$

In each row $b = tq$. The complete required file `check_examples.g` constructs these actual permutation groups. It verifies that the original group has order $n!$, checks its involution and string conditions, and tests every one of its sixty-four parabolic pairs. It also computes the actual subgroup intersection and the claimed permutation action. The resulting orders are

n	$\lvert B\rvert$	$\lvert H\rvert$	$\lvert K\rvert$	$\lvert H \cap K\rvert$	signs of (a, b, c)
9	12	288	40320	72	$(-, +, -)$
10	12	1440	768	24	$(-, -, +)$
13	12	1440	39916800	72	$(+, +, +)$.

These are finite mathematical premises for these three examples only; the common quantified proofs do not use sample computations.

For $n = 9$, K fixes 1 and acts as S_8 on $\{2, \dots, 9\}$. It is primitive there. The element $(2\,3\,4)$ belongs to $H \cap K$ but not to B. Thus the four-point case can fail without its imprimitivity condition.

For $n = 10$, the action of K on $\{3, \dots, 10\}$ preserves the nontrivial block system

$$\{3, 10\}, \quad \{4, 9\}, \quad \{5, 8\}, \quad \{6, 7\}.$$

The program verifies that this is exactly the orbit of the first displayed pair under K. But a, b are odd, and the pure transposition $(1\,2)$ is in $H \cap K$ and not in B. Thus the parity condition cannot simply be omitted from the five-point assertion.

For $n = 13$, all three marks a, b, c are even, but the projection of K to $\{3, \dots, 13\}$ is the primitive natural S_{11} action. The even cycle $(3\,4\,5)$ is in $H \cap K$ but not in B. Here the remaining assumption that fails is imprimitivity. The full permutation group K is consistent with evenness: an odd permutation in that eleven-point projection has the compensating transposition on the other pair of points. These verified examples recover the source's motivating conclusions without identifying projection parity with parity of the whole permutation.

4 Scope, retained inputs and verification

The operation and valid cases are due to De Saedeleer, Leemans and Mulpas. We use the full common paper, the complete eighteen-point correction proof, and the entire required example program. All are included in the proof boundary. No cost of common theory is silently borrowed from a different topic. The standard McMullen–Schulte end criterion is explicitly retained. The source's quoted rank-reduction result is background, already treated in the separate corrected L037 application; it is not needed as a premise of the positive augmentation proof. Neither is its quoted sesqui-extension result.

The closing sentence on source physical page 18 also needs a bond-index correction. On the distinguished t, a-orbit, ta is a three-cycle in the four-point row and a four-cycle in the other rows; on its complement it acts as a, of order at most two. Hence the first bond $|ta|$ is three or six in the four-point row, and four in the five- and six-point rows. The orbit bound always makes the *second* bond $|ab|$ three or six, as proved by the common dihedral calculation. It does not force the first bond to have those two values.

The numerical Table 1 on source physical page 2 reports a motivating computer experiment on $S_5, \dots, S_{11}$. Those census entries are retained background data, not regenerated or claimed as a new result here. The broader extensions suggested in its conclusion, to larger middle orbits and to higher ranks, remain outside this theorem. The generic bounded diagnostic `check_models.g` tests small instances of the common theorem, including actual subgroup identities; it is a validation file, not a proof of the unbounded statements. There is no new conjecture or numerical compression claim in this draft.

References

[1] J. De Saedeleer, D. Leemans and J. Mulpas, *A rank augmentation theorem for rank 3 string C-group representations of the symmetric groups*, arXiv:2212.12563v1, 23 December 2022.